

**EMERGENCY VEHICLE DETECTION AND
TRIGGERING SIGNAL IN
SUMO SIMULATOR**

Group 25-26J-473

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in
Software Engineering

Department of Computer Science & Software Engineering

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Declaration page of the candidates & supervisor

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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(Ms. Karthiga Rajendran)

Date.....

Abstract

One of the critical challenges in Sri Lankan cities is traffic congestion at intersections, which leads to excessive waiting times and inefficient traffic flow. Traditional fixed-time traffic signals cannot adapt to the highly dynamic and heterogeneous nature of local traffic, which includes a mix of motorbikes, three-wheelers, cars, vans, lorries, and buses. These inefficiencies become especially problematic when emergency vehicles or public transport require priority passage.

This research proposes the development of an intelligent real-time **traffic signal timing and priority control agent** integrated into the Simulation of Urban Mobility (SUMO) platform. Unlike traditional approaches, the system dynamically calculates and adjusts green and red durations based on live traffic demand, emergency events, and vehicle categories. A special focus is given to **emergency vehicle clearance**, where the agent guarantees a green corridor within **5–15 seconds** of detection, and **bus prioritization**, recognizing the vital role of public transport in Sri Lanka.

The agent operates by processing inputs from external detection modules (e.g., visual/audio emergency detection) and SUMO traffic state data, applying adaptive algorithms such as **queue-based pressure logic** and **reinforcement learning policies**. Through TraCI (Traffic Control Interface), the agent directly modifies signal phases, ensuring optimized flow while maintaining safety constraints and fairness across approaches.

By prioritizing both emergency and public transport vehicles, the system not only improves critical response times but also enhances person-throughput and reduces delays for the wider traffic stream. Simulation experiments in SUMO will demonstrate the effectiveness of this approach under realistic Sri Lankan traffic scenarios, paving the way for scalable and intelligent traffic management solutions.

Keywords – Traffic Signal Control, Intelligent Agent, SUMO, TraCI, Reinforcement Learning, Emergency Vehicle Priority, Bus Signal Priority, Adaptive Traffic Management

List Of Abbreviations

SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
RL	Reinforcement Learning
MARL	Multi-Agent Reinforcement Learning
ML	Machine Learning
AI	Artificial Intelligence
PPO	Proximal Policy Optimization (RL algorithm)
DQN	Deep Q-Network (RL algorithm)
DDPG	Deep Deterministic Policy Gradient (RL algorithm)
KPI	Key Performance Indicator
API	Application Programming Interface

Table 1 : List of Abbreviations

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1. Introduction

1.1. Background and Literature survey

Traffic congestion at urban intersections remains a critical challenge in Sri Lanka, where a diverse mix of vehicles—ranging from motorcycles and three-wheelers to buses and lorries—competes for limited road space. Traditional **fixed-time traffic signals** dominate the current system, yet these signals cannot adapt to rapidly changing demand, resulting in long queues, inefficient flow, and excessive waiting times. Such inefficiencies are particularly harmful in scenarios where **emergency vehicles** or **public transport** require immediate passage. Delays in clearing ambulances and fire trucks can lead to life-threatening consequences, while buses carrying large volumes of passengers experience unfair delays that reduce the overall efficiency of the transport network [1]

Existing research highlights that conventional traffic signal strategies are inadequate for heterogeneous, high-density traffic environments typical of South Asia [2] While some countries have implemented vehicle-actuated and adaptive signal control, Sri Lanka still heavily relies on pre-timed plans or even manual police intervention during peak hours [3]. These approaches lack consistency, scalability, and fairness, creating a pressing need for automated, intelligent systems that can balance efficiency with priority handling.

This research addresses this gap by focusing on the **development of an intelligent traffic signal timing and priority control agent**. The proposed system leverages the **Simulation of Urban Mobility (SUMO)** environment to model intersections and uses the **Traffic Control Interface (TraCI)** to dynamically adjust signal phases. The agent calculates phase durations in real time based on traffic state, queue lengths, and vehicle types, while enforcing special rules for emergency vehicles (ensuring green clearance within 5–15 seconds of detection) and for buses, which are prioritized due to their significance in Sri Lanka's public transport system. By simulating these control strategies in SUMO, the study aims to demonstrate

measurable reductions in delay, improved emergency response times, and enhanced person-throughput compared to traditional methods.

Advantages of the Proposed System

The proposed intelligent signal timing and priority agent offers several advantages over traditional fixed-time or manually controlled traffic signals:

- **Adaptive Signal Timing** – Dynamically adjusts green and red durations based on real-time demand, reducing unnecessary waiting times.
- **Emergency Vehicle Pre-emption** – Guarantees clearance for ambulances, fire trucks, and police vehicles within **5–15 seconds**, improving critical response times.
- **Public Transport Priority** – Incorporates **bus prioritization** to increase person-throughput and support sustainable mobility in Sri Lanka
- **Fair and Safe Control** – Balances major and minor road traffic, avoiding the starvation problems seen in manual police-based control.
- **Simulation-Driven Evaluation** – Integration with **SUMO and TraCI** enables rigorous testing under realistic traffic scenarios before deployment in real intersections.

1.2. Research Gap

Traditional methods of traffic signal control in Sri Lanka have mostly been **human-based** or **fixed-time scheduled**. Manual intervention by traffic police requires continuous human energy and often depends on driver cooperation. This leads to high risks of accidents, inconsistent decision-making, and unfair prioritization of major roads over minor ones. Fixed-time signals, while structured, cannot adapt to daily traffic fluctuations, causing unnecessary waiting times and congestion during peak hours.

Meanwhile, upcoming **Vehicle-to-Vehicle (V2V) communication-based approaches** are promising but impractical for the Sri Lankan context, as they require every vehicle to be equipped with specialized hardware, which is neither cost-effective nor scalable. These systems also operate independently of traffic signals, failing to directly reduce waiting times at intersections.

Existing **vehicle-actuated and adaptive control systems** have been explored in other regions, but in Sri Lanka they remain largely unimplemented. Furthermore, most prior research has not addressed the dual need for **emergency vehicle clearance (within 5–15 seconds)** and **public transport priority**, both of which are critical for safety and efficiency.

To address these gaps, the proposed research formulates **traffic signal control as a Reinforcement Learning (RL) problem**. By integrating with SUMO via TraCI, the RL-based agent will dynamically calculate signal timings based on real-time conditions, prioritize buses to maximize person-throughput, and ensure rapid corridor clearance for emergency vehicles.

Table 2 shows the comparison of traffic signal control approaches that have been investigated in prior research. Each method demonstrates certain strengths but also suffers from critical limitations when applied to Sri Lankan traffic conditions. Fixed-time and manual systems fail to adapt to real-time variations, actuated systems require costly infrastructure, and adaptive/AI approaches have not yet been localized to heterogeneous traffic environments. To overcome these limitations and achieve improved performance, the proposed system formulates traffic signal control as a **Reinforcement Learning (RL) problem**, enabling the agent to dynamically adjust timings, ensure 5–15 second clearance for emergency vehicles, and prioritize public transport within the SUMO simulation environment.

Method	Strengths	Research Gaps	References
Fixed-Time Signals	Simple, predictable, low-cost	✗ Cannot adapt to daily variations ✗ Long queues during peaks ✗ No priority for emergency/buses	[5] [6]
Manual Police Control	Flexible in oversaturation; can clear major queues	✗ Labor-intensive ✗ Inconsistent decisions ✗ Neglects side streets; safety risks	[7] [8]
Vehicle-Actuated Signals	Real-time responsiveness via detectors	✗ Requires costly infrastructure ✗ Limited adoption in Sri Lanka ✗ No built-in emergency handling	[9] [10]
Adaptive/AI-Based Control	Dynamic optimization (ML, fuzzy, rule-based); improves average flow	✗ Limited local research ✗ Requires sensor/data reliability ✗ Rarely tuned for mixed traffic	[11] [12] [13]
Emergency Priority Prototypes	RFID/sensor-based fast clearance for ambulances	✗ Only single-junction solutions ✗ No corridor-level	[14] [15]

		coordination ✗ Ignores public transport	
Proposed RL Agent	Learns optimal policies; ensures 5–15s emergency clearance ; prioritizes buses; adapts to heterogeneous demand	✓ Bridges all gaps by unifying adaptive control, emergency pre-emption, and bus priority in SUMO/TraCI	

Table 2 : Competitive Analysis

1.3. Research Problem

In modern urban environments, the effectiveness of traffic management systems is heavily influenced by their ability to **adapt signal timings dynamically** to real-time conditions and grant priority passage to critical vehicles such as ambulances, fire trucks, and buses. Traditional traffic signal systems in Sri Lanka operate primarily on **fixed timers** or rely on **manual police intervention**, both of which lack the capability to respond intelligently to unpredictable traffic surges or emergency events. While some research has explored **vehicle-actuated signals**, these require costly sensor infrastructure and still fail to provide coordinated emergency clearance or public transport prioritization.

Existing adaptive approaches that employ optimization or AI techniques remain rare in Sri Lanka and are typically designed for **homogeneous traffic** environments, ignoring the heterogeneous mix of motorcycles, three-wheelers, cars, and buses. Furthermore, current research seldom integrates **real-time adaptive control** with traffic simulation frameworks like **SUMO**, which are essential for evaluating system scalability and performance under complex, urban traffic conditions.

A critical gap is the **absence of Reinforcement Learning (RL)-based signal control** tailored for Sri Lankan traffic. RL provides the ability to model each intersection as an intelligent agent that learns optimal policies by interacting with its environment. However, most global RL studies overlook key requirements such as:

- Guaranteeing **5–15 second emergency vehicle clearance**,
- Providing **bus signal priority** to improve person-throughput, and
- Ensuring fairness so that non-priority vehicles are not starved.

The proposed research addresses this problem by formulating traffic signal timing as a **Reinforcement Learning problem** within the SUMO simulation framework, enabling the intelligent agent to dynamically minimize delays while ensuring safety-critical emergency and public transport prioritization.

2. Objectives

2.1. Main Objective

To design and develop an **intelligent traffic signal timing and priority control agent** integrated with the SUMO traffic simulator that dynamically adjusts signal phases in real time to minimize waiting times, guarantee **5–15 second clearance for emergency vehicles**, and provide **bus/public transport prioritization**, thereby improving efficiency, safety, and fairness in Sri Lankan urban traffic environments.

2.2. Specific Objectives

- **Formulate Traffic Signal Control as a Reinforcement Learning Problem** – Define the state space (queues, waiting times, vehicle types), action space (phase switching/extension), and reward function (delay minimization, emergency clearance, bus priority).
- **Develop Adaptive Signal Timing Algorithms** – Implement baseline logic such as pressure-based control and extend with RL-based policies to optimize green/red durations dynamically.

- **Integrate with SUMO and TraCI** – Use TraCI API to apply the agent’s decisions to intersections in SUMO, enabling real-time adaptive signal control and emergency pre-emption.
- **Emergency Vehicle Pre-emption** – Guarantee **5–15 second clearance** by detecting incoming emergency events (from teammate’s detection module) and immediately adjusting signal phases.
- **Bus and Public Transport Priority** – Incorporate priority weighting for buses to improve person-throughput and support sustainable urban mobility.
- **Evaluate System Performance** – Measure metrics such as average waiting time, queue lengths, emergency clearance time, bus delays, and fairness across approaches under varying Sri Lankan traffic conditions.
- **Visualization & Reporting** – Provide dashboards and logs showing signal states, priority events, queue lengths, and key performance indicators for analysis and validation.

3. Methodology

The proposed methodology focuses on developing an **intelligent traffic signal timing and priority control agent** that dynamically calculates signal phases in real time. The system integrates with the SUMO simulation environment through the TraCI API and applies **Reinforcement Learning (RL)** to minimize delays, guarantee emergency clearance, and prioritize buses/public transport. The approach is divided into key phases as follows:

3.1. Requirement Gathering

Requirement gathering for this component involves collecting, organizing, and preparing all necessary specifications and traffic data to ensure realistic simulation of Sri Lankan traffic conditions. This includes defining system requirements, intersection models, vehicle categories, and priority rules.

3.1.1. Dataset & Traffic Scenario Collection

To model realistic traffic and evaluate system performance, a multi-source data acquisition strategy will be followed:

1. **Traffic Flow Data** – Collect real intersection traffic counts (vehicle volumes, mix of cars, bikes, buses, three-wheelers, lorries) from Sri Lankan urban roads.
2. **Emergency Event Scenarios** – Generate synthetic scenarios in SUMO for ambulances and fire trucks approaching intersections at peak and off-peak times.
3. **Public Transport Data** – Integrate bus routes and schedules into SUMO to test person-throughput priority.
4. **Simulation Environment** – Use NetEdit to design and configure intersections representative of Colombo, Kandy, and Galle urban layouts.

3.2. Development Methodology

3.2.1. RL-Based Signal Timing Agent

Description: Formulate the traffic signal control problem as a Reinforcement Learning task:

- **State:** Queue lengths, waiting times, current phase, vehicle categories (emergency, bus, car, bike, etc.).
- **Action:** Switch to next phase, extend current phase, or terminate early.
- **Reward:** Negative of total weighted waiting time, heavy penalty for emergency delays, bonus for bus throughput.

Technologies & Tools:

- **Python 3.x** – Core development.
- **Stable-Baselines3** – RL algorithms (DQN, PPO, DDPG).
- **Gymnasium** – RL environment wrapper for SUMO.
- **NumPy/Pandas** – Feature extraction and state representation.

3.2.2. Baseline Control (Pressure/Actuated)

Description: Implement queue-pressure based logic and semi-actuated timing as a benchmark. These algorithms will serve as a comparison against the RL-based agent.

Technologies & Tools:

- Python, TraCI API for SUMO.
- Queue length sensors (lane area detectors in SUMO).

3.2.3. SUMO and TraCI Integration

Description: Connect the RL agent and baseline algorithms to the SUMO traffic simulation environment using TraCI. The system dynamically modifies signal phases and records performance metrics.

Technologies & Tools:

- **SUMO** (Simulation of Urban Mobility) – Traffic simulation software.
- **TraCI (Traffic Control Interface)** – Python API for controlling signal phases.
- **NetEdit** – SUMO network editor for creating intersections.

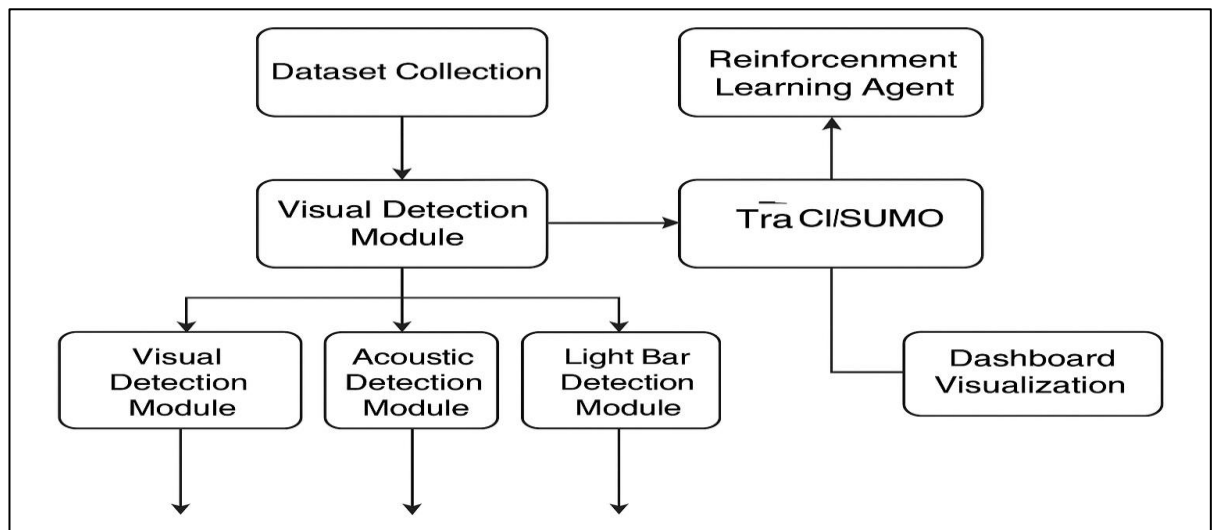


Figure 1 : Reinforcement Learning Framework for Traffic Signal Control

3.2.4. Emergency Pre-emption Module

Description: Upon receiving emergency detection input (from teammate's module), the agent guarantees a **5–15 second clearance** by pre-empting current phases and creating a green corridor.

Technologies & Tools:

- TraCI commands for emergency phase switching.
- Rule-based safety layer (min yellow/all-red constraints).

Figure 2 is the virtual environment where the traffic-based programs can be done it is more flexible and portable to use.

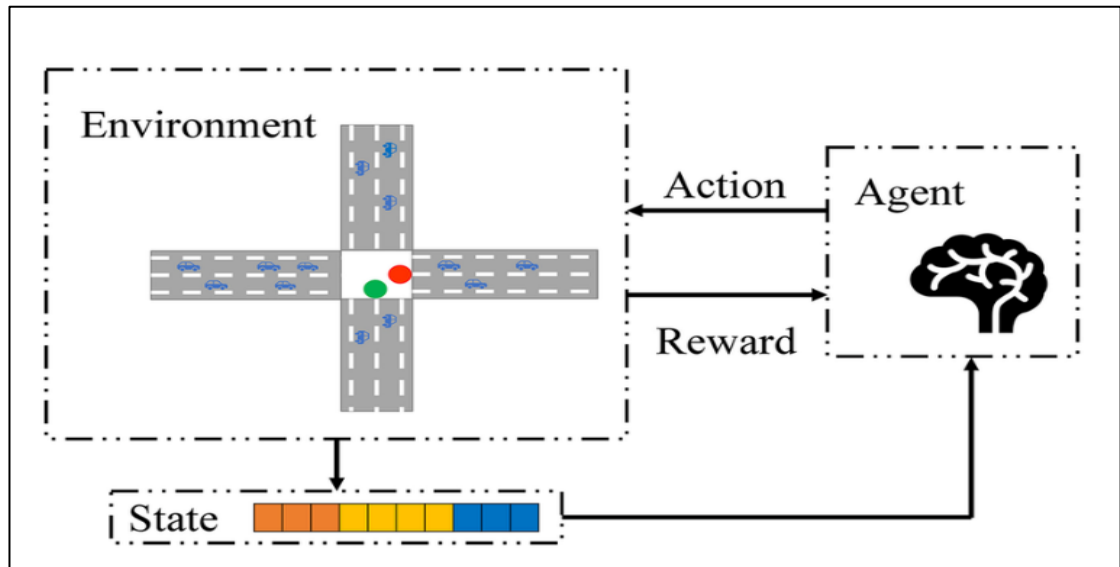


Figure 2 : SUMO Simulation Integrated with Reinforcement Learning Agent

3.2.5. Bus & Public Transport Priority

Description: Implement priority weighting for buses in the reward function, ensuring reduced delays for high-capacity vehicles while balancing fairness.

Technologies & Tools:

- SUMO bus route simulation.
- RL reward shaping for person-throughput optimization.

3.3. Project Management Methodology

Research requires effective project management, where scope, time, and cost must be carefully aligned through a structured system or procedure. For the adaptive quiz platform project, a process-oriented methodology will be employed, dividing the project into distinct stages with meticulously planned inputs and outputs. This approach is designed to accommodate changes, identify risks, and manage the project effectively. JIRA will be used to support this methodology, providing a robust platform for tracking progress, managing tasks, and ensuring that project requirements and timelines are met efficiently.

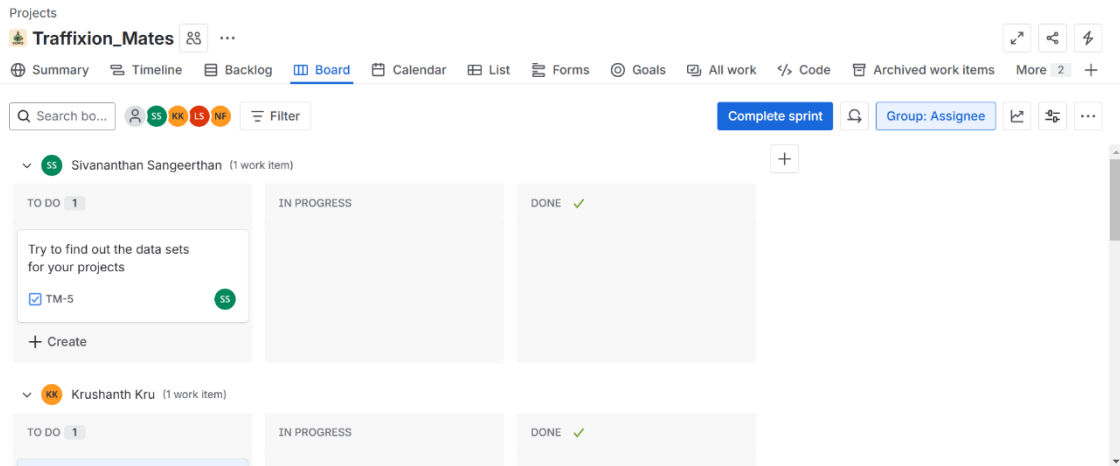


Figure 3 : Jira management tool

3.4. Feasibility Study

The feasibility study assesses the practicality of implementing the proposed Intelligent Traffic Signal Timing and Priority Control Agent in terms of technical capabilities, project scheduling, and economic viability.

3.4.1. Technical Feasibility

The proposed system integrates traffic simulation, reinforcement learning, adaptive control algorithms, and real-time priority handling for emergency and public transport vehicles. The following technical resources and competencies are available within the team to ensure successful development:

Knowledge on Technologies

To develop the proposed solution, the knowledge in below mentioned technologies should be wanted.

- **Traffic Simulation:** Hands-on experience with SUMO and the TraCI API for real-time signal control and data collection.
- **Reinforcement Learning (RL):** Proficiency in formulating state, action, and reward spaces for traffic control agents, and training using frameworks like **Stable-Baselines3** and **Gymnasium**.
- **Adaptive Control Algorithms:** Understanding of **queue-based pressure models** and semi-actuated control logic for benchmarking against RL.
- **Machine Learning Tools:** Skills in Python, NumPy, Pandas, and PyTorch/TensorFlow for RL model training, fine-tuning, and evaluation.
- **Backend & Visualization:** Experience with FastAPI/Flask for simulation data serving and React.js/Plotly for dashboard visualization of traffic signal states and performance metrics.
- **Data Processing:** Ability to collect, preprocess, and simulate traffic flows that reflect Sri Lankan traffic characteristics (vehicle mix, bus routes, emergency events).

Knowledge on Tools

To develop the proposed optimization model, all members should have an in-depth understanding of development tools, simulation platforms, and project management

frameworks. Jira will be used to facilitate project tracking, sprint planning, and task collaboration throughout the development process.

- **Development Tools:** VS Code, Jupyter Notebook, Google Colab (GPU-enabled), Git & GitHub
- **Traffic Simulation Tools:** SUMO, TraCI API, NetEdit
- **Machine Learning / RL Tools:** Stable-Baselines3, Gymnasium, PyTorch, TensorFlow, NumPy, Pandas
- **Project Management:** Jira for sprint tracking and collaboration
- **Dashboard Tools:** Flask/FastAPI, React.js, Plotly/Chart.js/D3.js, Matplotlib

Data Collection & Preprocessing Skills

- Ability to collect datasets from Google Images, Kaggle, and real-world recordings.
- Experience in image augmentation, audio denoising, and annotation to improve model robustness.

3.4.2. Schedule Feasibility

The project will follow the Agile methodology, divided into 2–3 week sprints for each module:

- Traffic scenario preparation (Sprint 1)
- Baseline control logic (Sprint 2)
- Reinforcement Learning formulation (Sprint 3)
- RL agent development (Sprint 4)
- SUMO & TraCI integration (Sprint 5)
- Emergency vehicle pre-emption (Sprint 6)
- Bus/public transport priority (Sprint 7)

- Dashboard development (Sprint 8)
- Testing & optimization (Sprint 9)

Given the modular approach, the timeline is realistic and allows flexibility for refinement based on test feedback. Milestones will be tracked in **Jira**, ensuring tasks are delivered within deadlines.

3.5. System Analysis

3.5.1. Software Solution Approach

The proposed system will be developed as a modular, reinforcement learning-based signal control platform that integrates SUMO traffic simulation with intelligent decision-making agents. Unlike detection modules (handled by teammates), this component focuses on calculating and optimizing signal timings in real time to reduce delays and ensure emergency and bus priority.

Traffic Scenario Setup – Create Sri Lankan-style intersections in SUMO (via NetEdit) with a heterogeneous vehicle mix (motorbikes, three-wheelers, cars, vans, buses, lorries, emergency vehicles).

Baseline Control – Implement fixed-time and queue-pressure control algorithms to serve as benchmarks for comparison.

RL Agent Training – Define state (queues, waiting time, emergency events, bus presence), action (phase switching/extension), and reward (minimize total delay, heavy penalty for emergency waiting, bonus for bus throughput). Train using Stable-Baselines3 with SUMO+Gym integration.

Emergency Vehicle Pre-emption – Integrate signals from the detection module; guarantee **5–15s clearance** by dynamically pre-empting current phases and coordinating adjacent intersections.

Bus/Public Transport Priority – Apply weighted reward functions to reduce bus delays and increase person-throughput.

SUMO Integration – Use TraCI API for real-time control of intersection signal phases and extraction of traffic metrics.

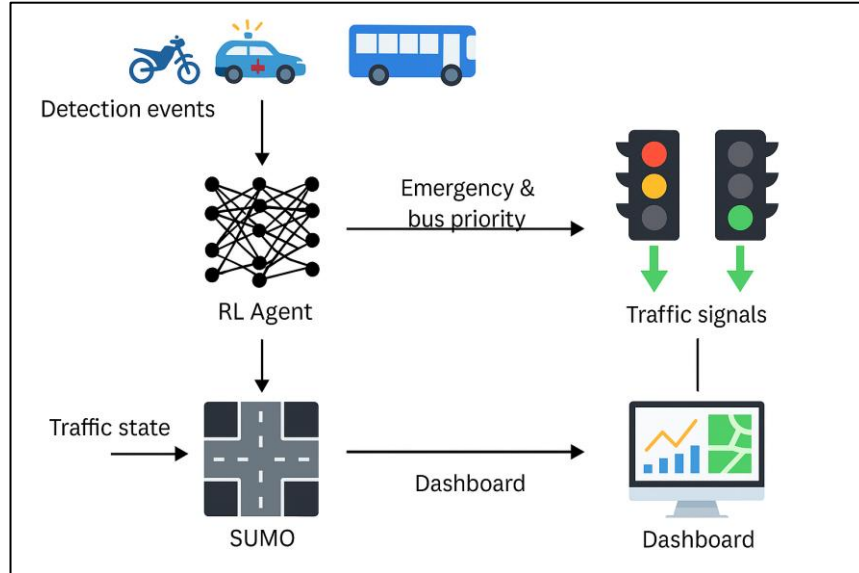


Figure 4 : System Diagram

3.5.2. Tools and Technologies

The proposed system will be developed using a combination of programming languages, simulation platforms, reinforcement learning frameworks, and visualization tools to ensure robustness, scalability, and real-time performance.

Programming Languages

- **Python 3.x** – Core development language for reinforcement learning models, SUMO integration, and traffic control logic due to its extensive ecosystem of ML and simulation libraries.
- **JavaScript (React.js)** – For building the real-time, interactive dashboard frontend.

Frameworks and Libraries

- **Stable-Baselines3 / Gymnasium** – Reinforcement Learning frameworks for training and evaluating signal timing agents.
- **PyTorch / TensorFlow** – Deep learning frameworks for implementing RL models and custom training functions.
- **NumPy / Pandas** – For data handling, preprocessing, and statistical analysis of traffic states and logs.
- **Matplotlib / Seaborn / Plotly** – For visualizing performance metrics, training progress, and evaluation charts.

Traffic Simulation and Integration Tools

- **SUMO (Simulation of Urban Mobility)** – Microscopic, open-source traffic simulator for testing adaptive signal control strategies in Sri Lankan traffic scenarios.
- **TraCI API** – Python interface for controlling SUMO simulations in real time (signal switching, detector monitoring).
- **NetEdit** – SUMO's network editor for creating and modifying traffic network layouts (urban intersections, corridors).

Backend Development and API Tools

- **Flask / FastAPI** – Python-based web frameworks for creating RESTful APIs and WebSocket services to serve traffic signal states and simulation data to the dashboard.

Development and Project Management Tools

- **VS Code** – Main IDE for coding and debugging.

- **Google Colab / Jupyter Notebook** – For RL experiments and GPU-enabled training.
- **Jira** – For Agile sprint planning, backlog management, and task tracking.
- **Git & GitHub** – For version control and collaborative development.

Testing Tools

- **Pytest** – For automated unit and integration testing of Python modules.
- **Postman** – For testing and validating REST API endpoints.

Hardware Requirements

- **GPU-enabled Workstation / Cloud GPU** – For training RL models efficiently (PPO, DQN, DDPG).
- **Standard PC/Laptop** – For running SUMO simulations, agent training, and dashboard.
- **Optional Sensors/Inputs (for future real-world extension)** – Integration with external detection events (from teammate's modules) via APIs.

3.6. Project Requirements

The project requirements define the expected functionality and operational qualities of the **Intelligent Traffic Signal Timing and Priority Control Agent**. These requirements ensure that the system achieves both technical performance goals and real-world applicability within Sri Lankan traffic conditions.

3.6.1. Functional Requirements

The system shall:

- **Adaptive Signal Timing** – Dynamically calculate and adjust green/red durations in real time based on traffic demand, queue lengths, and vehicle mix.
- **Reinforcement Learning Integration** – Formulate and train an RL agent with state, action, and reward definitions for optimal traffic signal control.
- **Emergency Vehicle Pre-emption** – Guarantee **5–15 second clearance** by prioritizing emergency vehicles upon detection and coordinating across adjacent intersections.
- **Bus and Public Transport Priority** – Apply weighted signal priority for buses to improve person-throughput while maintaining fairness.
- **Baseline Algorithms** – Implement fixed-time and queue-pressure control for benchmarking against the RL agent.
- **SUMO Integration** – Interface with SUMO through TraCI API for real-time traffic state extraction and signal control.
- **Performance Evaluation** – Collect simulation data to evaluate metrics such as waiting time, queue length, clearance time, and fairness.
- **Dashboard Visualization** – Display real-time simulation outputs (signal states, queues, priority events, and KPIs) through a web-based dashboard.
- **Data Logging** – Store traffic state logs, agent decisions, and performance metrics for analysis and validation.

- **Scalable Architecture** – Enable extension to multiple intersections (multi-agent setup) for corridor-level traffic optimization.

3.6.2. Non-Functional Requirements

The system shall:

- **Performance** – Operate with decision latency of less than **1 second per simulation step** in SUMO.
- **Effectiveness** – Reduce average waiting time by at least **20%** compared to fixed-time signals; clear emergency vehicles within **≤15s**.
- **Scalability** – Support integration with additional intersections and larger SUMO networks with minimal modifications.
- **Reliability** – Maintain consistent control logic across varying traffic densities, incidents, and peak/off-peak scenarios.
- **Security** – Secure simulation logs and dashboard communication using encrypted data transfer.
- **Maintainability** – Use modular code structure with well-documented RL agent, SUMO interface, and dashboard modules.
- **Usability** – Provide an intuitive dashboard interface accessible to traffic engineers without advanced technical knowledge.
- **Compatibility** – Run on both **local mid-range PCs** and **cloud-based GPU environments** for RL training.
- **Resource Efficiency** – Optimize RL computations and SUMO integration to avoid excessive CPU/GPU usage during long simulations.

3.7. Project Scope

The scope of this project defines the boundaries and extent of the **Intelligent Traffic Signal Timing and Priority Control Agent** component, outlining the features and functionalities that will be developed and those that will not be included in the current phase.

3.7.1. In-Scope

The system will include:

Adaptive Signal Timing – Dynamically adjust green and red durations in real time based on traffic demand, queue lengths, and heterogeneous vehicle categories (motorbikes, cars, buses, lorries, etc.).

Reinforcement Learning Agent – Formulate and train an RL agent to optimize signal control, minimizing delays while ensuring fairness and efficiency.

Baseline Algorithms – Implement fixed-time and pressure-based control strategies for benchmarking against the RL agent.

Emergency Vehicle Pre-emption – Guarantee **5–15 second clearance** for ambulances, fire trucks, and police vehicles by pre-empting current phases and creating green corridors.

Bus and Public Transport Priority – Apply weighted prioritization for buses and high-capacity vehicles to enhance person-throughput and promote sustainable urban mobility.

SUMO Integration – Connect the RL agent with the SUMO simulator via the TraCI API for real-time control of intersections.

Multi-Intersection Coordination – Share state and priority information across neighboring intersections to avoid bottlenecks and support corridor-level optimization.

Dashboard Visualization – Provide a real-time web-based dashboard (React.js + FastAPI) to display traffic signal states, emergency events, queue lengths, and performance metrics.

Testing in Simulated Environments – Evaluate the system under diverse traffic conditions including peak/off-peak demand, emergency events, high bus volumes, and mixed traffic flows typical of Sri Lanka.

3.7.2. Out-of-Scope

The system will not include:

Real-World Deployment – The current project will be limited to SUMO simulations and will not control actual physical traffic lights at intersections.

Integration with V2X Communication Systems – Vehicle-to-everything communication (V2V, V2I) is excluded in this phase but may be explored in future extensions.

Full City-Wide Deployment – The system will be tested on selected intersection and corridor simulations in SUMO, not on an entire city-scale network.

Complex Incident Management – Accident detection, lane closures, and rerouting strategies will not be handled within this phase.

Integration with Live City Traffic Databases – Only synthetic and simulation-based data (traffic demand, routes, emergency events) will be used for testing, without integration into real-world traffic databases.

Driver Interaction Features – Gamification, driver alerts, or user training modules will not be developed.

3.8. Testing

Testing is a critical phase in the development of the **Intelligent Signal Timing and Priority Control Agent** to ensure system reliability, adaptability, and real-time

performance. Multiple testing approaches will be applied to validate individual modules, integrated functionality, and overall simulation outcomes.

3.8.1. Unit Testing

Purpose:

To validate that each independent module of the system works as intended in isolation before integration.

Scope:

- **Baseline Algorithms:** Verify fixed-time and pressure-based control modules produce expected signal phase transitions.
- **RL Agent:** Validate that the agent receives correct state inputs, produces valid actions, and updates reward values properly.
- **Emergency Pre-emption Logic:** Ensure the module correctly enforces **5–15s clearance** upon emergency event input.
- **Bus Priority Module:** Confirm weighted prioritization reduces bus waiting time without breaking fairness constraints.
- **Dashboard Components:** Check backend APIs and frontend visualizations update correctly with simulation data.

Tools:

- Pytest for automated testing of Python modules.
- Mock SUMO detectors and traffic scenarios for controlled test cases.

3.8.2. Integration Testing

Purpose:

To verify that all modules work together correctly when combined into the complete system.

Scope:

- Integration of RL agent outputs with SUMO via TraCI API for real-time control.
- End-to-end flow: detection module (from teammate) → emergency signal → pre-emption logic → SUMO update.
- Multi-intersection coordination (sharing emergency/bus priority info between intersections).
- Real-time update flow from backend (FastAPI) to dashboard (React.js).

Tools:

- Postman for API testing.
- Real-time SUMO logs and simulation playback for verification.

3.8.3. Performance Testing

Purpose:

To assess the system's responsiveness, decision quality, and computational efficiency.

Metrics:

- **Decision Latency:** Time taken for the RL agent to sense → decide → update signals (target < 1s per SUMO step).
- **Average Waiting Time:** Reduction percentage compared to fixed-time baseline.
- **Emergency Clearance Time:** Time taken to clear ambulances (goal: ≤ 15 s).
- **Bus Delay Reduction:** Average improvement in bus travel time vs baseline.
- **Fairness:** Longest waiting time of non-priority vehicles.

- **Resource Utilization:** Monitor CPU/GPU usage during training and SUMO integration.

Tools:

- Python profiling and logging utilities.
- Matplotlib/Seaborn for performance visualization.
- GPU/CPU monitoring tools during RL training.

4. Expected Outcomes

The **Intelligent Signal Timing and Priority Control Agent** is expected to deliver measurable improvements in overall traffic efficiency, emergency clearance, and bus prioritization through adaptive signal control and reinforcement learning. The outcomes will focus on performance, usability, and impact within simulated Sri Lankan traffic environments.

Measurable Performance Goals

Reduced Waiting Times

- Achieve at least **20% reduction** in average vehicle waiting time compared to fixed-time signals.
- Reduce **95th percentile delays** for vehicles in congested conditions.

Emergency Vehicle Pre-emption

- Guarantee **clearance of emergency vehicles within 5–15 seconds** of detection.
- Demonstrate consistent creation of green corridors in SUMO for ambulances and fire trucks.

Bus/Public Transport Priority

- Reduce bus delays by $\geq 15\%$ during peak periods.
- Increase overall **person-throughput** (passenger-based metric) compared to vehicle-only throughput.

Reinforcement Learning Efficiency

- Train an RL agent capable of adapting to heterogeneous Sri Lankan traffic.
- Ensure decision-making latency of **< 1 second per SUMO step**.

Scalability

- Demonstrate functional control across **multi-intersection networks**, coordinating priority events without bottlenecks.

5. Budget and Budget Justification

The cost of development will remain relatively low, as the system primarily uses **open-source simulation frameworks (SUMO, TraCI, Stable-Baselines3)** and programming libraries. The main expenses will involve **GPU-enabled computing for RL training, internet/hosting services** for the dashboard, and standard project documentation.

Estimated Budget:

Type	Cost (LKR)	Remarks
GPU-enabled computing (Cloud/Colab Pro)	8,000	Training model fine-tuning
Data Collection (camera, microphone)	20,000	Real-world capturing
Internets and hosting	6,000	Hosting dashboard
Stationery and documentation	1,000	Printing and report costs
Total	35,000	-

Table 3 : Budget and Budget Justification

The **low-cost structure**, combined with the scalability of the system, makes the project economically viable. The solution can later be extended, commercialized, or integrated into existing **Intelligent Transportation Systems (ITS)** for real-world deployment.

6. Timeline

The project will proceed through a well-organized development timeline. At the outset, the team will concentrate on completing the platform's design, including creating a comprehensive system diagram and choosing the essential technologies. Next, the implementation phase will commence, focusing on fine-tuning and developing algorithms for SUMO Stimulation. Simultaneously, performance analysis tools and other system components will be integrated. The final phase will include extensive testing, performance assessment, and the formulation of commercialization plans. Throughout these stages, the project will emphasize careful development and validation before the platform's official launch.

The components require data and model training with fine tuning process in the case the Figure 8 shows the time allocation for the whole year form the beginning.

	2025						2026				
Task	July	August	September	October	November	December	January	February	March	April	May
Feasibility study	Blue										
Evaluate feasibility and background study	Blue										
Requirement gathering	Blue										
Equipment gathering and testing		Green									
Background survey		Green									
Literature review		Green									
Requirement analysis		Green									
Software requirement specification		Green									
Functional, non-functional requirement		Green									
Proposal presentation		Green									
Project proposal report		Green									
Data collection		Green									
ML model creation			Orange								
Model training			Orange								
Model tuning			Pink	Pink	Pink						
Designing wireframes					Blue	Blue	Blue				
SUMO and Software integration					Blue	Blue	Blue				
Deployment & maintenance					Blue	Blue	Blue				
Progress presentation 1						Orange					
Research paper						Orange	Orange	Orange			
Progress presentation 2							Orange	Orange	Orange		
Testing									Orange	Orange	
Final presentation and viva											Green

Figure 5 : Component Timeline

7. Communication Management Plan

The Communications Management Plan is designed to ensure that all team members, as well as the supervisor and co-supervisor, receive the necessary information to effectively perform their roles throughout the project. Successful project execution relies on strategic planning and effective communication. This plan outlines how to communicate with various stakeholders efficiently, detailing the audience, content, format, frequency, and expected outcomes of communications. It also specifies each stakeholder's role, task assignment, and communication strategy based on their influence, interests, and expectations.

Communication Objectives

Proactive communication is essential for project success. Communication should be:

- Adequate: Providing information in the right format and content.
- Specific: Tailored to the targeted audience.
- Sufficient: Including all necessary details.
- Concise: Clear and to the point, avoiding unnecessary repetition.
- Timely: Addressing relevant points at appropriate times.

Communication Media The following media will be utilized for communication throughout the project:

1. Microsoft Teams : For formal updates, documentation, and detailed communications.
2. Documents (MS Word and PowerPoint): For reports, presentations, and detailed project documentation.
3. Phone Calls: For quick discussions and urgent matters.
4. Meetings: Conducted via meeting rooms, conference phones, or Microsoft Teams for in-depth discussions, planning, and progress reviews.
5. Chats (WhatsApp): For informal, quick exchanges and real-time updates.

8. Commercialization

The **Intelligent Signal Timing and Priority Control Agent** has strong potential for commercialization in both the **public sector** and **private sector** markets. The system addresses a critical urban challenge by reducing traffic congestion, ensuring rapid emergency clearance, and improving bus/public transport efficiency, making it valuable for deployment in **Intelligent Transportation Systems (ITS)** across Sri Lanka and globally.

Potential Markets

- Government Traffic Management Authorities
 - City councils and transportation departments can adopt the system to upgrade existing fixed-time signals to adaptive, RL-driven control, thereby improving network efficiency and emergency response times.
- Emergency Services
 - Ambulance, fire, and police departments can benefit from guaranteed **5–15 second clearance**, ensuring faster and safer passage through congested intersections.
- Public Transport Operators
 - Bus services can deploy the system to reduce delays during peak hours, improving schedule adherence and passenger satisfaction.
- Smart City Solution Providers
 - Technology vendors offering IoT-based traffic management solutions can integrate the RL-based control agent into their platforms as a scalable, adaptive module.

Future Commercial Potential

The technology can be extended beyond its current scope to support:

- **Multi-Intersection / Corridor Optimization:** Coordinated control of entire corridors or city grids, reducing bottlenecks across large areas.

- **Public Transport Priority Expansion:** Integration with bus tracking systems for real-time schedule-based priority.
- **Autonomous Vehicle Integration:** Providing adaptive signal priority for connected and driverless vehicles.
- **Green Mobility Support:** Optimizing signals for reduced idling, leading to lower emissions and supporting sustainability goals.

9. Summary

The proposed **Intelligent Signal Timing and Priority Control Agent** is an adaptive traffic management solution designed to reduce delays, guarantee rapid emergency clearance, and improve public transport efficiency in Sri Lankan urban environments. By formulating traffic signal control as a **Reinforcement Learning (RL) problem**, the system dynamically adjusts signal timings based on real-time traffic states, emergency events, and bus prioritization. Integration with **SUMO and TraCI** enables rigorous simulation, allowing for performance evaluation under diverse traffic conditions.

The project follows an **Agile development methodology**, ensuring iterative improvements through unit, integration, simulation, and performance testing. Expected outcomes include **reduced average waiting times, emergency vehicle clearance within 5–15 seconds**, improved **bus service reliability**, and a scalable foundation for multi-intersection and corridor-level optimization.

With strong commercialization potential in **government traffic authorities, emergency services, public transport operators, and smart city solution providers**, this system offers a **cost-effective, high-impact contribution** to modern urban mobility management. Its adaptability ensures long-term relevance, paving the way for real-world deployment and integration into future **Intelligent Transportation Systems (ITS)**.

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